

CAPSTONE PROJECT SUBMISSION

AI/ML Fundamentals Course, Module 8

Full Name

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Project Track

Field-Based Scenario Track. EDU-02: Student Performance Early-Warning Model (EdTech)

Project Title

Student Performance Early-Warning Model: predicting academic risk from early-course data (OULAD)

Repository URL

<https://github.com/Mubina-lazy/edu02-early-warning-model>

Short Project Description

This project is an early-warning classifier for online courses. After roughly the first 25 percent of a course has elapsed, it estimates whether a student is at risk of failing or withdrawing, so that academic advisors can intervene while there is still time to help. The model uses only information available at that point: demographics, virtual learning environment (VLE) click activity, and scores from assignments that were already due. I built it on the Open University Learning Analytics Dataset (OULAD, CC-BY 4.0), which covers 32,593 student enrolments across 22 course presentations.

Preventing data leakage was the central design constraint. I completed a documented data audit with an issue log, fixed the prediction point at 25 percent of each course presentation's length, and excluded every feature dated after that point, including final exam results, later assignment scores and whole-course activity totals. I also excluded students who had already unregistered by the cutoff, because their outcome is known at prediction time and keeping them would inflate the results. The data is split chronologically by course presentation: I train on the 2013 cohorts, validate on 2014B, and evaluate once on the unseen 2014J cohort.

I compared seven runs in MLflow: a majority-class baseline, logistic regression with and without class weighting, two random forest configurations and two XGBoost configurations, tuning the decision threshold for each. The final model is XGBoost at a threshold of 0.327, selected on validation. On the held-out 2014J cohort it reaches recall 0.760, precision 0.583, F1 0.660 and PR-AUC 0.714 for the at-risk class, against 0.000 recall for the majority baseline. Because missing a struggling student costs more than an unnecessary check-in, recall and F1 are my primary metrics rather than accuracy. The evaluation report also covers per-module results (including one course with no training data), fairness slices by disability, socio-economic band, gender and age, and an analysis of the at-risk students the model fails to catch.

Demo / Application Link

Reproducible Google Colab demo: `demo.ipynb` in the repository root. Open it in Colab and run all cells. No dataset download is required, because the trained model artifact (under 1 MB) is committed to the repository. The notebook runs two real unseen students, shows how invalid input is rejected, and covers a zero-activity edge case.

Access Instructions

The repository is public, so no special access is required. All code, evaluation reports, the trained model artifact and the Colab demo are available directly from the link above.

Use of AI assistance: in line with the Module 8 rules, I used an AI coding assistant (Claude) for scaffolding, refactoring and documentation support. I reviewed and tested every change, made the modelling and evaluation decisions myself, and can explain and defend each component of this submission.